


# F78PB Probability and Statistics B

## Hypothesis Testing

# Relationships between Two Variables

- We have data on two variables $X$ and $Y$ for a sample of $n$ subjects (or objects).
- Correlation and regression investigate a possible relationship between $X$ and $Y$.
- Examples:
  - 'Before' and 'after' measurements
  - Exam marks in different courses
  - Dexterity and anxiety when performing a task
  - Advertising and sales
- **Correlation:** measures how strong a relationship is.
- **Regression:** establishes a formula for the relationship so it can be used for prediction.
- Consider paired data $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$.
- First step: draw a scatterplot to see if there is any relationship and what form it might take.

# Notation

$$s_{xx} = \sum_{i=1}^{n}(x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - \frac{1}{n}(\sum_{i=1}^{n} x_i)^2$$

$$s_{yy} = \sum_{i=1}^{n}(y_i - \bar{y})^2 = \sum_{i=1}^{n} y_i^2 - \frac{1}{n}(\sum_{i=1}^{n} y_i)^2$$

$$s_{xy} = \sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^{n} x_i y_i - \frac{1}{n}\sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i$$

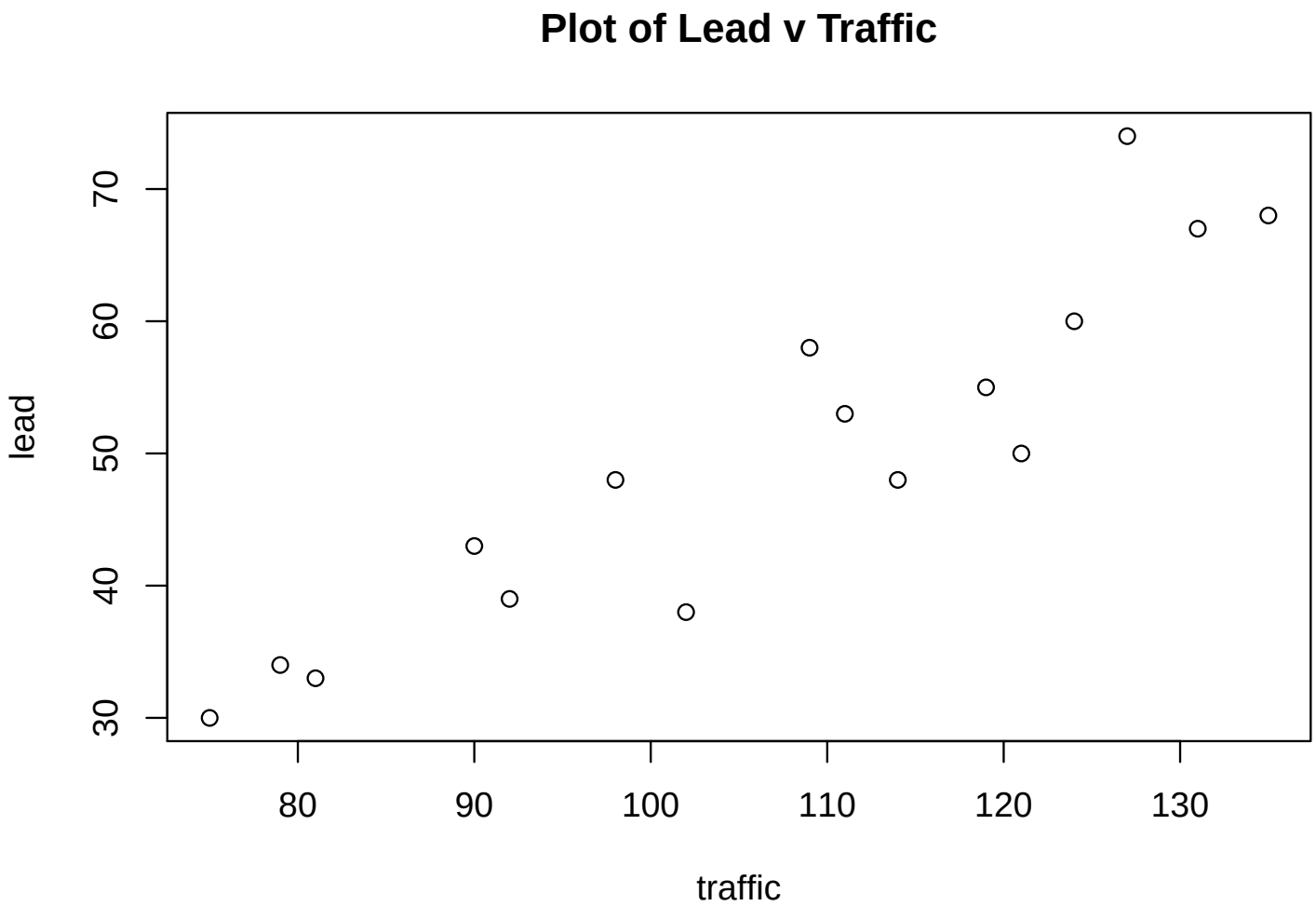
# Traffic Pollution Data Table

In an investigation of traffic pollution, information was collected from 16 busy roads. The data below are the amount of lead (in parts per million) in the ground level vegetation and the volume of traffic (hundreds of vehicles recorded per day):

| Road | Lead $(y_i)$ | Traffic $(x_i)$ | Road | Lead $(y_i)$ | Traffic $(x_i)$ |
|---|---|---|---|---|---|
| 1 | 34 | 79 | 9 | 50 | 121 |
| 2 | 58 | 109 | 10 | 55 | 119 |
| 3 | 43 | 90 | 11 | 67 | 131 |
| 4 | 60 | 124 | 12 | 68 | 135 |
| 5 | 53 | 111 | 13 | 30 | 75 |
| 6 | 48 | 114 | 14 | 39 | 92 |
| 7 | 74 | 127 | 15 | 33 | 81 |
| 8 | 38 | 102 | 16 | 48 | 98 |

Table: Traffic pollution dataset used for correlation and regression example.

# Example: Traffic Pollution




Scatterplot suggests positive linear relationship.

Now we formalise this by calculating the correlation coefficient $r$.

The **population** correlation coefficient is defined as

$$\rho = \frac{\mathbb{C}\text{ov}(X, Y)}{\sigma_x \sigma_y}$$

and the **sample** correlation coefficient is given by

$$r = \frac{s_{xy}}{\sqrt{s_{xx} s_{yy}}}$$

Technically $r$ is called the sample product-moment correlation coefficient or the Pearson correlation coefficient(there are others e.g. based on ranks).

# Interpretation of $r$

Recall that $r$ must lie in the range $-1 < r < +1$;

- $r = +1$ means that the points lie exactly on a line with positive slope;
- $r = -1$ means that the points lie exactly on a line with negative slope;
- $r$ being large means that the points lie close to a straight line; smaller values of $r$ mean that the points are more spread out;
- $r$ near zero means that there is no linear relationship for these points.

Notice that $r$ does **not** indicate 'cause and effect'.

For our example:

$$\sum x = 1708, \sum x^2 = 187890,$$
$$\sum y = 798, \sum y^2 = 42454, \sum xy = 88714$$

$$s_{xy} = 88714 - \frac{1}{16}(1708)(798) = 3527.5$$
$$s_{xx} = 187890 - \frac{1}{16}(1708)^2 = 5561$$
$$s_{yy} = 42454 - \frac{1}{16}(798)^2 = 2653.75$$

So

$$r = \frac{s_{xy}}{\sqrt{s_{xx}s_{yy}}} = \frac{3527.5}{\sqrt{5561 \times 2653.75}} = 0.918$$

This is large and positive as expected from the scatterplot.

# Normal model and inference for $\rho$

Under the assumption that $(x, y)$ **has the bivariate normal distribution**, we can test

$$H_0 : \rho = \rho_0 \quad \text{versus} \quad H_1 : \rho \neq \rho_0$$

using the approximation

$W = \frac{1}{2} \log \frac{1+r}{1-r} \sim N(\frac{1}{2} \log \frac{1+\rho}{1-\rho}, \frac{1}{n-3})$

The same approximate sampling distribution may be used for confidence intervals for $\rho$.

## Example

The sample correlation coefficient between two samples is found to be $r = 0.936$ $(n = 10)$. Test the hypothesis that there is no correlation between the two corresponding populations at the $1\%$ significance level.

# Linear Regression

- We often believe that one variable $Y$ depends on another variable $X$, and that we can predict $Y$ for a given $X$.
- Here:
  - $Y$: dependent (response) variable
  - $X$: independent (explanatory) variable
- Example: In the lead pollution study:
  - $Y$ = amount of lead in vegetation
  - $X$ = volume of traffic
- Regression concerns fitting a function to represent this relationship.
- We consider linear regression, where we fit a straight line to $(x, y)$ data.
- Note: Values of $X$ are controlled by the experimenter and are not random.

# Model for Regression

- The regression model:

$$Y_i = \alpha + \beta x_i + \varepsilon_i, \quad i = 1, 2, \ldots, n$$

  where:
  - $e_i$ are independent error terms
  - Assume that $\mathbb{E}[e_i] = 0$ and let $\sigma^2 = \mathbb{V}\text{ar}(e_i)$ (Often we assume $e_i \sim N(0, \sigma^2)$)
- This model has three parameters:
  - $\beta$: slope coefficient
  - $\alpha$: intercept (constant)
  - $\sigma^2$: error variance
- The error variance accounts for variation in the data about the straight line.
- Note that the regression model can also be written as:

$$\mathbb{E}(Y_i) = \alpha + \beta x_i, \quad i = 1, 2, \ldots, n$$

$$\mathbb{V}\text{ar}(Y_i) = \sigma^2$$

$$\mathbb{C}\text{ov}(Y_i, Y_j) = \mathbb{C}\text{ov}(\varepsilon_i, \varepsilon_j) = 0, i \neq j \qquad \text{(independent observations)}.$$

# Fitting the Regression Line

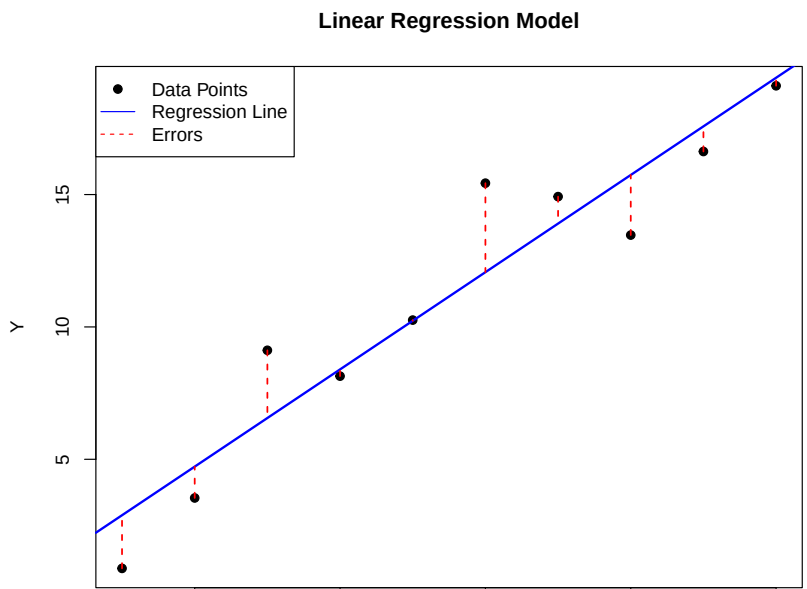
- We fit the line:

$$Y_i = \alpha + \beta x_i + \varepsilon_i$$

  using the method of least squares for data $(x_1, y_1), \ldots, (x_n, y_n)$.
- Goal: choose $\hat{\alpha}$ and $\hat{\beta}$ to minimize the error sum of squares:

$$S(\alpha, \beta) = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2, \quad \hat{y}_i = \alpha + \beta x_i$$

- Graphically: the fitted line minimizes the sum of squared vertical deviations from the data points.



# Fitted Regression Line

- Least Squares minimizes:

$$S(\alpha, \beta) = \sum_{i=1}^{n} \big(y_i - (\alpha + \beta x_i)\big)^2$$

- To find $\hat{\alpha}, \hat{\beta}$, set derivatives to zero:

$$\frac{\partial S}{\partial \alpha} = 0, \quad \frac{\partial S}{\partial \beta} = 0$$

- From $\partial S/\partial \alpha = 0$:

$$\sum_{i=1}^{n} y_i = n\hat{\alpha} + \hat{\beta} \sum_{i=1}^{n} x_i$$

  Rearranged:

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$$

  (So the fitted line passes through $(\bar{x}, \bar{y})$.)

# Estimating the Slope

- Least Squares minimizes:

$$S(\alpha, \beta) = \sum_{i=1}^{n} \big(y_i - (\alpha + \beta x_i)\big)^2$$

- From $\partial S/\partial \beta = 0$:

$$\sum_{i=1}^{n} x_i y_i = \hat{\alpha} \sum_{i=1}^{n} x_i + \hat{\beta} \sum_{i=1}^{n} x_i^2$$

- Substitute $\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$ and simplify:

$$\hat{\beta} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{s_{xy}}{s_{xx}}$$

- Then:

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$$

# Summary: LS Estimates and Properties

- **Least Squares Estimates** for our experiment $(x_1, y_1), \ldots, (x_n, y_n)$

$$\hat{\beta} = \frac{s_{xy}}{s_{xx}}, \quad \hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}.$$

- **Least Squares Estimators** for a random experiment $(x_1, Y_1), \ldots, (x_n, Y_n)$

$$\hat{\beta}(\underline{x}, \underline{Y}) = \frac{s_{xY}}{s_{xx}}, \quad \hat{\alpha}(\underline{x}, \underline{Y}) = \bar{Y} - \hat{\beta}(\underline{x}, \underline{Y})\bar{x}, \quad \hat{y}_i(\underline{x}, \underline{Y}) = \hat{\alpha}(\underline{x}, \underline{Y}) + \hat{\beta}(\underline{x}, \underline{Y})x_i.$$

- **Properties of $\hat{B}$:**
  1. $\mathbb{E}(\hat{\beta}(\underline{x}, \underline{Y})) = \beta$ (unbiased)
  2. $\mathbb{V}\text{ar}(\hat{\beta}(\underline{x}, \underline{Y})) = \dfrac{\sigma^2}{s_{xx}}$
- **Error Variance Estimate**: Define

$$\hat{\sigma}^2 = \frac{1}{n-2}\sum_{i=1}^{n}(Y_i - \hat{y}_i)^2 = \frac{1}{n-2}\sum_{i=1}^{n}(Y_i - \hat{\alpha} - \hat{\beta}x_i)^2$$

Then $\hat{\sigma}^2(\underline{x}, \underline{Y})$ is unbiased.